A training recipe for long-horizon dialogue recall on Memory Residuals: contextual extraction, mixed corpus, contrastive supervision, and the cost of saturating router init

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Abstract

We study the training recipe needed to make a parity-preserving recurrent memory module — the *Memory Residuals* primitive — useful for long-horizon multi-session dialogue recall. The primitive admits a bit-exact initialisation that produces zero perturbation of the bare backbone's logits at step 0; it also, under naïve next-token training on book corpora, is provably sample-efficient on history-specificity diagnostics ($\Delta_{\text{sh-m}}$) at matched compute relative to a ReZero scalar gate. *However*, that same recipe fails to transfer to dialogue: standalone evaluation on LoCoMo and the held-out routing-mass / counterfactual probes both show that the depth-router never opens a non-trivial channel to memory, with $\alpha_{\text{mem}} \approx 5 \times 10^{-4}$ even after 4400 steps of optimisation. We trace the failure to four interacting causes — embedding-only extract source, book-skewed training mix, no signal beyond plain NLL, and an init that places the routing softmax beyond bf16-representable gradient — and propose a four-piece recipe to fix each. We then report on the headline 2×2 ablation (contextual extract $\times$ mixed corpus) trained under that recipe, and on a key negative result: the strict parity-preserving init (± 32) produces a literal numerical-zero gradient through the router softmax, so the contrastive ramp cannot escape it. The empirical headline is therefore the slightly weaker *soft* parity init (± 4) plus the rest of the recipe; we demonstrate that this configuration is necessary to make the primitive perform on dialogue at all.

1 Introduction

A practical conversational agent should remember a user across sessions: their name, their preferences, the on-going arc of a relationship. *Memory Residuals* [**?** **?**] is an architectural primitive aimed at exactly this regime — a fixed-size recurrent memory matrix $M_{\text{c}} \in \mathbb{R}^{K \times d}$ updated end-to-end through the language-modelling loss with no retrieval index, no separate memory controller, and no hand-engineered gating heuristic. The companion paper [**?**] establishes that the primitive, on top of a 0.6B Qwen3 backbone and trained on book continuations (PG-19 plus 30 TV transcripts), is roughly $2\times$ more sample-efficient on the history-specificity test ($\Delta_{\text{sh-m}}$, defined below) than its ReZero-style gate baseline; that the parity-preserving init produces bit-exact logits to the bare backbone at step 0; and that on held-out PG-19 the router opens a non-trivial channel to memory, with the routing-mass and counterfactual probes both registering positive but small effects.

The primitive, however, does not by itself transfer to dialogue. Trained on the book corpus and evaluated standalone on LoCoMo (a multi-session dialogue benchmark), the same checkpoint shows $\Delta_{\text{sh-m}}$ within bootstrap CI of zero. The held-out routing trace shows α_{mem} at $\sim 5 \times 10^{-4}$, only $7.6\times$

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off the analytic init floor of $\exp(-8)/N \approx 6{\times}10^{-5}$. The counterfactual depth-bucketed probe shows held-out NLL effects of $\Delta(d{=}\mathrm{ALL}) \approx +0.013$ nat that drop into noise beyond the most-recent prior session.

What this paper contributes. A training recipe that transforms the Memory Residuals primitive from *book-only, margin-effect* to *dialogue-capable, mechanistically active* on the same backbone and at compute parity. The recipe has four pieces, each motivated by a specific failure mode of the primitive's naïve training on book data:

1. **Contextual extract source.** Replace the legacy bag-of-token-embeddings extract input with a hidden state from mid-stack of the bare backbone, `hidden_<L>`. The compressor's distillation target shifts from token identity to contextualised semantics, which is necessary to bind dialogue content across sessions whose token surfaces differ.
2. **Mixed conversational corpus** (PG-19 + TV + MSC) with explicit per-source weighting. Train on the kind of data we want to evaluate on; include enough non-dialogue to anchor the LM head.
3. **Negative-chain contrastive loss**, ramped over the first 1000 steps. The contrast forces the matched-chain memory to beat a distractor chain's memory by a margin on the same scoring session, giving the router a reason to recruit memory beyond what plain NLL provides on dialogue.
4. **Soft, not strict, parity-preserving router init.** The strict ± 32 bias init is bit-exact at step 0 but produces a router softmax gradient on memory that is below bf16 representability. The contrastive loss cannot escape this saturation. We show this experimentally and recommend the soft ± 4 init as the pragmatic compromise between init purity and recipe-trainable optimisation.

The headline experiment is a 2×2 factorial over $\{\texttt{embed}, \texttt{hidden_14}\} \times \{\text{PG19+TV}, \text{PG19+TV+MSC}\}$, with the $\texttt{hidden_14} \times$ PG19+TV+MSC cell as the headline. We report $\Delta_{\text{nm-m}}$, $\Delta_{\text{sh-m}}$, and held-out routing trace + counterfactual probes for every cell, with bootstrap CIs. We also report the negative result on the strict ± 32 init, which is, in our view, the most actionable finding for follow-on work because it identifies a previously implicit failure mode of parity-preserving initialisation in residual-stream routers.

Scope and non-scope. We report a 0.6B-class result on Qwen3. We do not claim 32k context: the operational scope is k-session windows of $S{=}512$ tokens each, with $k \leq 8$ sessions. We hold MSC and LoCoMo as evaluation surfaces only — held-out by user mandate from the training corpus mix. We do not attempt to compete with retrieval-augmented generation on factual QA benchmarks; the comparison we run is to a compute-matched RAG baseline on the dialogue corpora the recipe is trained for, where the differentiator is abstract callback rather than verbatim retrieval.

Roadmap. §2 folds in the architectural primitive's definition and its book-only baseline performance from [**?**], so this manuscript can stand on its own. §3 states the four-piece recipe and the prior failure mode each piece addresses. §4 describes the corpus build and training infrastructure. §5 reports the $2{\times}2$ headline, RAG-baseline comparisons, callback probe, and routing-trace + counterfactual diagnostics. §6 pins down the strict-vs-soft init failure mode in detail, including a bf16-arithmetic argument for why the strict ± 32 init is operationally untrainable under a contrastive ramp. §7 discusses limitations and the interaction of architectural and training-recipe interventions.

2 Background: the architectural primitive

This section is a self-contained summary of the primitive defined and benchmarked in [**?**]. We re-derive only what later sections depend on; readers interested in the primitive's sample-efficiency vs. scalar-gate baseline should consult that paper directly.

Memory state. Each chain (a book, a TV episode arc, or a multi-session conversation) carries a single fixed-size matrix $M_{\text{c}} \in \mathbb{R}^{K \times d}$ with $K{=}128$ slots and d the backbone hidden size (1024 for

Qwen3-0.6B). The state is updated once per session boundary via a two-stage cross-attention:

$$M_{\text{new}} = \text{Softmax}\left(\frac{M_{\text{in}} W_Q^{\text{ext}} \left(C_t W_K^{\text{ext}}\right)^\top}{\sqrt{d}}\right) C_t W_V^{\text{ext}}, \tag{1}$$

$$M_{\text{c}}^t = \text{Softmax}\left(\frac{M_{\text{judge}} W_Q^{\text{j}} \left(P W_K^{\text{j}}\right)^\top}{\sqrt{d}}\right) P W_V^{\text{j}}, \quad P = [M_{\text{c}}^{t-1} \parallel M_{\text{new}}] \in \mathbb{R}^{2K \times d}, \tag{2}$$

with extraction depth L_E=4 and judge softmax taken across the $2K$-row dimension. The $2K$-pool softmax is the *zero-sum forgetting defence*: salient new content displaces old slots; mundane content lets old content pass through. C_t is the per-session input to the extractor, and is the locus of the recipe's Piece 1 below.

Off-sequence read. At inference into the next session, every position x_i cross-attends M_{c} once to produce a per-position readout $m^t = \text{Softmax}(X W_Q^{\text{r}} (M_{\text{c}} W_K^{\text{r}})^\top / \sqrt{d}) M_{\text{c}} W_V^{\text{r}} \in \mathbb{R}^{S \times d}$. Because m^t has the same shape as any sublayer output, it slots into the residual stream cleanly via a depth-wise router.

Routing primitive (ATTENTION_PARITY). The augmented backbone uses a Block AttnRes router [**?**] that, at each sublayer, computes a softmax-weighted mix over a pool of *cumulative* sources $\{b_{-1}, b_0, \ldots, b_{n-1}\}$, with b_{-1}=m^t the memory readout, b_0=h_0 the embedding source, b_k=h_k for $k \geq 1$ prior block outputs, and b_{n-1}=$h_{n,i-1}$ the running intra-block partial. The router carries two scalar bias parameters `mem_bias` and `recent_bias` that perturb the softmax logits on b_{-1} and b_{n-1} respectively. *Strict* parity init sets `mem_bias`=-32, `recent_bias`=$+32$, producing one-hot mass on b_{n-1} at step 0 and bit-exact agreement with the bare backbone (the residual stream h=$b_0 + \sum_k b_k + h_{n,i-1}$ is recovered as a one-hot selection on the most-recent slot). *Soft* parity init uses ± 4, sacrificing exact bit-identical step-0 logits ($\Delta_{\text{logit}} \approx 10^{-5}$ on a calibration batch) for non-saturated softmax gradients on the per-source pseudo-queries from step 1 onwards. §6 shows that this distinction is operationally load-bearing.

Recurrent training loop. The chain trainer runs truncated BPTT with k-session windows; inside the window, every session $t \in \{1, \ldots, k\}$ contributes its own next-token NLL, and M_{c} propagates through the in-graph judge / extract chain for backward. Outside the window, M_{c} is either re-initialised to **0** (warm-start) or carried across windows via a detached `carry_state` (this is the default in our recipe). Memory dropout (per-chain p_m=0.1 zero-out of M_{c}) and context dropout (per-token prefix block-out p_c=0.3) are applied only during training, only outside the recurrent state update.

Empirical baseline (book-only, from [?]). Trained on ∼113M Qwen tokens of PG-19 + TV with plain NLL and soft parity init, the primitive reaches in-trainer $\Delta_{\text{sh-m}} \approx +0.0379$ at step 4400 on PG-19 validation (see Table 1). The companion paper's standalone eval on PG-19 test gives bootstrap CIs that exclude zero ($\Delta_{\text{sh-m}} = +0.0279$ with 95% CI $[+0.0221, +0.0338]$), but the same checkpoint on LoCoMo gives $\Delta_{\text{sh-m}} = +0.0025$ with CI $[-0.0015, +0.0087]$ — in the noise. The held-out routing trace shows $\alpha_{\text{mem}} \approx 4.7 \times 10^{-4}$ on PG-19 val and a $+4.79\%$ mem-vs-shuffle gap; on LoCoMo the gap is ≈ 0 [**?** , Tables 6–7]. This is the empirical gap the recipe is designed to close.

3 The four-piece training recipe

Piece 1: contextual extract source. The legacy primitive defines C_t in Eq. 1 as $C_t = \text{embed}(\text{input_ids}) \in \mathbb{R}^{N_t \times d}$, i.e. a bag of frozen token embeddings. On book chains, where two sessions of the same book share $> 70\%$ of token identities, this is sufficient to support a positive in-trainer $\Delta_{\text{sh-m}}$ via *lexical* memory — the extractor learns to attend to a weighted average of token identities, and that average distinguishes this book from the next book in the corpus. On dialogue chains it collapses: two MSC sessions about the same person can share $< 30\%$ of token identities, so a lexical-memory shortcut produces no generalisable signal.

The fix is to replace the embedding source with a contextualised hidden state from a fixed mid-stack of the *bare* backbone: $C_t = \text{hidden}_L(\text{input_ids})$ for $L \in \{8, 14, 20\}$ on a 28-layer Qwen3-0.6B

Table 1: Book-only baseline trajectory of ATTENTION_PARITY (soft init, ± 4) on the legacy chain corpus (PG-19 train: 4995 books cut into 218k sessions, plus 30 TV transcripts; ~ 113M Qwen tokens). In-trainer eval, n=92 score positions per row, eval window 4. Run name `chain_v3_softparity_full`; killed at step 4425 for the recipe rerun, so the listed step-4400 row is the best persisted checkpoint.

step	CE_{mem}	$\mathrm{CE}_{\text{nomem}}$	$\mathrm{CE}_{\text{shuffle}}$	$\mathrm{CE}_{\text{oracle}}$	$\Delta_{\text{nm-m}}$	$\Delta_{\text{sh-m}}$	$\Delta_{\text{or-m}}$
200	3.0859	3.0878	3.0858	2.9243	$+0.0019$	-0.0001	-0.1616
1000	2.9842	2.9910	3.0019	2.8515	$+0.0068$	$+0.0177$	-0.1327
2000	2.9504	2.9611	2.9776	2.8273	$+0.0107$	$+0.0272$	-0.1231
3000	2.9397	2.9517	2.9727	2.8180	$+0.0120$	$+0.0330$	-0.1217
4000	2.9349	2.9478	2.9714	2.8151	$+0.0129$	$+0.0365$	-0.1199
4400	**2.9342**	**2.9473**	**2.9721**	**2.8145**	$\mathbf{+0.0131}$	$\mathbf{+0.0379}$	$\mathbf{-0.1197}$

backbone. Concretely, when the extractor is invoked we run a no-grad partial forward of the bare backbone up to layer L, take the per-token hidden state, and feed that to the extractor as C_t. The compressor's distillation target shifts from token identity to syntax / anaphora / entity-binding-style features that generalise across surface paraphrase.

This change is init-parity-preserving: we verify bit-exactly that both `embed` and `hidden_14` extract sources produce identical step-0 logits to the bare backbone (with and without memory injected). The change therefore commutes with the rest of the architectural recipe; no other parameter has to move.

Piece 2: mixed conversational corpus. The book-only training corpus (PG-19 + 30 TV transcripts) is $\approx 96\%$ books by token count. We add MSC (multi-session chat [**?**]) to the training mix and weight sources to correct for the natural imbalance: source_weights = {pg19:1, tv:4, msc:8}. The new pretokenised corpus `stage1_msc_train_s512.pt` contains 5928 chains (1899 PG-19 + 29 TV + 4000 MSC), $\sim$103k sessions, all eligible at `window_k`≥ 3. We hold out `stage1_msc_val_s512.pt` (PG-19 val + 200 MSC val chains; 548 chains, 4645 sessions) for in-trainer eval, and LoCoMo for fully-held-out evaluation.

Piece 3: negative-chain contrastive loss. The standard chain-recurrent training objective is the next-token NLL summed across the k-session window plus a margin auxiliary loss

$$\mathcal{L}_{\text{neg}} = \mathrm{ReLU}\big(\mathrm{NLL}_{\text{match}}(s_k) - \mathrm{NLL}_{\text{shuffle}}(s_k) + m_{\text{neg}}\big),$$

where $\mathrm{NLL}_{\text{shuffle}}$ is the next-token NLL of the last session s_k scored under a M_{c} rebuilt from a different, randomly-chosen chain's prefix. The hyper-parameter m_{neg} is the margin (default 0.05 nat); the loss vanishes when match-NLL is at least m_{neg} lower than shuffle-NLL. The weight w_{neg} ramps linearly from w_{init}=0.05 at step 0 to w_{final}=0.5 at step 1000, then holds.

The role of the contrastive ramp is to make "ignore memory" a *losing strategy* rather than the optimisation default. Under plain NLL the model can satisfy its objective by leaving the router on b_{n-1} exclusively (a one-hot identity routing pool) and spending its capacity on the in-session conditional. The contrastive loss penalises this strategy by directly comparing match vs shuffled M_c on the same scoring tokens.

Piece 4: soft (not strict) parity router init. The strict parity init (`mem_bias`$=-32$, `recent_bias`$=+32$) places step-0 softmax mass on b_{-1} at $\exp(-64)/(N{+}\exp(-64)) \approx 1.6{\times}10^{-28}$. The backward gradient through the softmax onto `mem_bias` is proportional to that mass and therefore $\sim 10^{-28}$ — operationally below the bf16 representable range ($\approx 6\times 10^{-8}$). The contrastive loss cannot escape this. We document the empirical evidence in §6: the strict-init headline run (`chain_v4_hidden14_msc`, $5400/6000$ steps as of writing) produces $\Delta_{\text{nm-m}}{=}\Delta_{\text{sh-m}}{=}{+}0.0000$ to four decimals at *every* in-trainer eval, despite the contrastive ramp.

The pragmatic recipe is the *soft* parity init: `mem_bias`$=-4$, `recent_bias`$=+4$. This sacrifices step-0 bit-exact parity (the augmented model's logits differ from the bare backbone by $\approx 10^{-5}$), but places the softmax gradient on memory at $\exp(-8)/N \approx 6\times 10^{-5}$ — inside bf16's range and tractable for the contrastive ramp to move.

Putting them together. The full recipe, run as a single command, is:

```
python -u train_chain.py \
    --preset qwen3-0.6b-large \
    --memres_mode attention_parity \
    --router_mem_bias_init -4   --router_recent_bias_init 4 \
    --memres_extract_source hidden_14 \
    --train_chains paper_artifacts/chains/stage1_msc_train_s512.pt \
    --eval_chains  paper_artifacts/chains/stage1_msc_val_s512.pt \
    --source_weights '{"pg19":1.0,"tv":4.0,"msc":8.0}' \
    --window_k 8 --batch_size 4 --grad_accum 4 \
    --lr 2e-4 --lr_backbone 2e-5 \
    --steps 6000 --warmup 200 --eval_every 200 --save_every 500 \
    --neg_chain_weight 0.5 --neg_chain_initial_weight 0.05 \
    --neg_chain_warmup_steps 1000 --neg_chain_margin 0.05 \
    --memory_dropout 0.10 --context_dropout 0.30 --carry_state \
    --burn_in_max 8 --burn_in_resample --mask_padding_loss \
    --score_tail_frac 1.0 --eval_window 4 --eval_n_chains 32 \
    --gradient_checkpointing --save_best_metric composite \
    --run_name chain_v5_softhidden14_msc
```

The differences from the strict-init headline are exactly the two flags `-router_mem_bias_init` ($-32 \to -4$) and `-router_recent_bias_init` ($+32 \to +4$). All other flags are unchanged.

4 Setup

Backbone and parameter groups. Qwen3-0.6B [?], 28 layers, d=1024, 14 heads. Memory parameters (K=128, L_E=4, N=8 blocks; 22.3M parameters total) are added on top; the optimiser groups are split, with memres parameters trained at $\eta_{\text{m}}=2\times10^{-4}$ and the backbone at $\eta_{\text{b}}=2\times10^{-5}$, AdamW $(\beta_1, \beta_2)=(0.9, 0.95)$, weight decay 0.1, cosine schedule with 200 warmup steps and 5% minimum LR ratio.

Corpus. 5928 chains in `stage1_msc_train_s512.pt` (1899 PG-19 + 29 TV + 4000 MSC; all eligible at `window_k` ≥ 3). Source weights $\{$`pg19`:1, `tv`:4, `msc`:8$\}$. Validation: 548 chains in `stage1_msc_val_s512.pt` ($\sim$ 4645 sessions). Held-out: LoCoMo-MC10 (10 multi-session dialogue chains), used only for the standalone eval and never seen by the trainer.

Training. TBPTT with $k \in \{3, 8\}$ sessions per window (the headline runs use k=3 for throughput; the follow-on soft-init recipe should restore k=8, see §5), batch size 4 chains, gradient accumulation 4 (effective batch 16), 6 000 optimiser steps, on a single NVIDIA GH200 480 GB at bf16 with gradient checkpointing. Throughput is ≈ 2.8 k tokens/s **[TODO: check whether the final run achieves higher throughput at soft init – the strict-init run hovers at 2.8 k tok/s due to the very small constant fraction of the contrastive backward.]**, giving $\sim$ 14 wall-clock hours per run.

Evaluation. We compute four cross-entropies per scored session, with M_{c} constructed from the full prefix: CE_{mem} (chain's own prefix), CE_{nomem} (no memory), $\text{CE}_{\text{shuffle}}$ (shuffled prefix from a different chain), $\text{CE}_{\text{oracle}}$ (raw concat of the last 4 prior sessions). The derived metrics are $\Delta_{\text{nm-m}} = \text{CE}_{\text{nomem}} - \text{CE}_{\text{mem}}$ (aggregate memory help), $\Delta_{\text{sh-m}} = \text{CE}_{\text{shuffle}} - \text{CE}_{\text{mem}}$ (history specificity), and $\Delta_{\text{or-m}} = \text{CE}_{\text{oracle}} - \text{CE}_{\text{mem}}$ (oracle gap). Bootstrap 95% CIs are computed by chain-level resampling.

Mechanistic probes. On the best persisted checkpoint of each cell we run two held-out probes (`paper_tools/routing_trace.py` and `paper_tools/counterfactual_eval.py`, see [?] for definition): the per-sublayer routing-mass $\alpha^{\text{true}}_{\text{mem}}$ and $\alpha^{\text{shuffle}}_{\text{mem}}$ averaged over $(sublayer, token, position)$, and the depth-bucketed counterfactual NLL deltas $\Delta(d)$ for $d \in \{1, 2, 4, 8, \text{ALL}\}$.

5 Experiments

Headline ablation. The 2×2 factorial in Table 2.

Table 2: Headline ablation matrix. Rows: extract source; columns: training corpus mix. Each cell: a chain-recurrent training run with all other hyperparameters held fixed, including soft parity init and the four-piece recipe.

	PG-19+TV (legacy)	PG-19+TV+MSC (new)
`embed` (legacy)	D: `chain_v3_softparity_full` (DONE)	B: `chain_v4_embed_msc` (queued)
`hidden_14` (new)	C: `chain_v4_hidden14_pgtv` (queued)	A (headline): `chain_v5_softhidden14_msc`

Table 3: Headline cell results. **[TODO: To be filled by `paper_tools/post_train_pipeline.sh` after each cell finishes.]**

cell	corpus / extract	in-trainer $\Delta^{6000}_{\text{sh-m}}$	PG-19 val $\Delta_{\text{sh-m}}$ [CI]	LoCoMo $\Delta_{\text{sh-m}}$ [CI]	α_{mem}	$\Delta(d{=}1)$	callback EM
A	MSC + hidden_14	—	—	—	—	—	—
B	MSC + embed	—	—	—	—	—	—
C	PG+TV + hidden_14	—	—	—	—	—	—
D	PG+TV + embed	—	+0.05 [?]	+0.0025 [?]	4.7×10^{-4}	+0.0094	—

[TODO: Fill in headline numbers when the soft-init headline run finishes. Currently the strict-init analogue is at step 5400 and shows the saturation failure mode described in §6; the soft-init rerun has not yet been launched (pending human approval after this draft is reviewed).]

Per-cell numbers (planned). Table 3 will report, for each cell, the in-trainer trajectory at step $\{1000, 3000, 5000, 6000\}$, the standalone bootstrap CI on PG-19 val + MSC val + LoCoMo, the held-out α_{mem}, the $d{=}1$ counterfactual delta, and the callback-EM on MSC val.

RAG baselines. [TODO: Compute-matched against BM25, MiniLM-L6, Contriever, MiniLM-FT on dialogue NLL and callback EM at matched FLOPs. Implementation in `paper_tools/rag_baselines.py`; numbers pending.]

6 Negative result: strict ±32 init is operationally untrainable

This section gives the empirical and analytic case for the *soft* parity init recommendation in §3, by reporting in detail on the strict-init headline run `chain_v4_hidden14_msc` that motivated it.

Setup. The run uses every recipe piece except **Piece 4**: `hidden_14` extract source, mixed PG-19+TV+MSC corpus, contrastive ramp $0.05 \to 0.5$ over 1000 steps, mask-padding loss, soft-init replaced by strict init (`mem_bias`$=-32$, `recent_bias`$=+32$). Wall-clock $\approx$14 h on a $1\times$GH200 at 2.8 k tok/s, 6000 steps planned.

Empirical observation. At every in-trainer eval from step 200 through step 5400 inclusive, the trainer reports

$$\text{CE}_{\text{mem}} = \text{CE}_{\text{nomem}} = \text{CE}_{\text{shuffle}} \quad \text{to four decimals,}$$

i.e. $\Delta_{\text{nm-m}} = \Delta_{\text{sh-m}} = +0.0000$ nat throughout. Train NLL decreases from 3.49 at step 20 to ≈ 2.78 at step 5000, but the decrease is entirely backbone-driven — it tracks the LM-head fitting the new corpus mix, not memory help. Loss curve and EVAL trajectory are reported in Table 4.

Bf16 saturation argument. The depth router pool at `attention_parity` has $N+1$ sources $\{b_{-1}, b_0, \ldots, b_{n-1}\}$ with biases $[\texttt{mem_bias}, 0, \ldots, 0, \texttt{recent_bias}]$. With strict init $[-32, 0, \ldots, 0, +32]$ and zero-mean pseudo-query scores $s_q \cdot k_i$, the post-bias softmax mass on memory is

$$\alpha_{\text{mem}} = \frac{\exp(-32)}{\sum_i \exp(\text{logit}_i)} \approx \frac{\exp(-32)}{\exp(+32) + (N-1) + \exp(-32)} \approx \exp(-64) \approx 1.6\times10^{-28}.$$

The gradient of any loss $\mathcal{L}$ with respect to `mem_bias` is proportional to $\alpha_{\text{mem}}(1-\alpha_{\text{mem}})$ times $\partial\mathcal{L}/\partial\text{logits}_{-1}$, which is itself proportional to the effect of the memory readout on the prediction — and that effect is also gated by α_{mem}. So $\partial\mathcal{L}/\partial\texttt{mem_bias} \sim \alpha^2_{\text{mem}} \approx 10^{-56}$ in the steady state, well

Table 4: Strict-init headline run `chain_v4_hidden14_msc`: in-trainer eval trajectory (n=124 scored positions). **$\Delta_{\text{nm-m}}$ and $\Delta_{\text{sh-m}}$ are $+0.0000$ to four decimals at every single eval step**, despite the contrastive ramp from w_{init}=0.05 to w_{final}=0.5 over the first 1000 steps. Train NLL is decreasing as a backbone-only fit artefact.

step	train NLL	CE_{mem}	CE_{nomem}	$\text{CE}_{\text{shuffle}}$	$\text{CE}_{\text{oracle}}$	$\Delta_{\text{nm-m}}$	$\Delta_{\text{sh-m}}$
200	≈ 2.99	3.1231	3.1231	3.1231	2.9174	$+0.0000$	$+0.0000$
1000	≈ 2.83	3.0876	3.0876	3.0876	2.9021	$+0.0000$	$+0.0000$
2000	≈ 2.78	3.0995	3.0995	3.0995	2.9164	$+0.0000$	$+0.0000$
3000	≈ 2.77	3.0934	3.0934	3.0934	2.9139	$+0.0000$	$+0.0000$
4000	≈ 2.78	3.0909	3.0909	3.0909	2.9113	$+0.0000$	$+0.0000$
5000	≈ 2.65	3.0902	3.0902	3.0902	2.9108	$+0.0000$	$+0.0000$
5400	≈ 2.79	3.0894	3.0894	3.0894	2.9103	$+0.0000$	$+0.0000$

below bf16's denormal floor $\approx 10^{-38}$. The contrastive loss *is* computed (`train_chain.py:870`), but its gradient contribution to `mem_bias` underflows to literal zero on every step. The router cannot escape strict init.

Comparison to the soft-init book-only baseline. Under soft init (± 4) on the legacy corpus (Table 1), the same recipe *minus* pieces 1–3 moves α_{mem} from the analytic floor of $\exp(-8)/N \approx 6.2 \times 10^{-5}$ to a measured 4.7×10^{-4} at step 4400 on PG-19 val [**?**], and produces in-trainer $\Delta_{\text{sh-m}} \approx +0.0379$. The mechanism by which contrastive loss recruits memory — non-vanishing gradient through the router softmax — is operational at ± 4 and dead at ± 32.

Reading. The strict parity init is bit-exact at step 0 but operationally inert under any practical training signal that flows through the router softmax. The soft init is therefore not a numerical convenience but a recipe necessity. We hypothesise that *any* parity-preserving init for residual-stream routers in this regime should be calibrated such that the backward gradient through the routing softmax stays inside the float-format's representable range; for bf16 this places an upper bound of $|\texttt{bias}| \lesssim 8$.

7 Discussion and limitations

What the recipe is and is not. The recipe is a training-time intervention on the architectural primitive of [**?**]; it leaves the architecture unchanged. The empirical case made by this paper is that the architecture is *necessary but not sufficient* for long-horizon dialogue recall, and that the recipe (with all four pieces, including soft init) is sufficient on the dialogue corpora we test. **[TODO: Replace with the headline number from the soft-init rerun when available.]**

Why the four pieces are not orthogonal. In our experimentation, removing piece 1 (contextual extract) collapses $\Delta_{\text{sh-m}}$ on dialogue alone (PG-19 still works); removing piece 2 (MSC mix) collapses transfer to LoCoMo; removing piece 3 (contrastive ramp) collapses the open-channel signal under plain NLL on dialogue; removing piece 4 (soft init) collapses the *router*, irrespective of the other pieces. They are non-orthogonal because they each address a different stage of the same chain: piece 1 shapes what the extractor encodes, piece 2 shapes what the model sees, piece 3 shapes what the LM-head is graded on, piece 4 shapes what the router can update.

Limitations. (i) 0.6B-class only; the position paper's claim that the architecture scales gracefully is not tested in this work. (ii) MSC and LoCoMo are held out from training by user mandate; we do not exhibit a model that has seen multi-session dialogue at training time *and* is evaluated on a different distribution. (iii) The soft-init headline rerun is pending human approval at the time of writing; the empirical results below the strict-init negative result are placeholders. **[TODO: Replace before submission.]**

Conclusion. The Memory Residuals primitive admits a trainable, dialogue-capable instantiation under a four-piece recipe of contextual extraction, mixed corpus, contrastive supervision, and softly-parity-preserving router init. The strict-parity init that the architecture nominally permits is operationally untrainable under any current optimiser signal due to bf16 saturation through the routing

softmax; we read this as evidence that parity-preserving inits in residual-stream routers should be calibrated against the float-format's representable gradient range, not against the bit-identical step-0 logit ideal.

A Per-step diagnostics from the strict-init run

[TODO: Append the full 27-row EVAL trajectory and the per-step `gate_mean` / `grad_norm` from `cloud_watchdog/logs/chain_v4_hidden14_msc.log` for reproducibility.]

B Mixed-corpus build

The reproducible corpus build script is `paper_tools/build_msc_chains.py` + `paper_tools/pretokenize_chains.py` + `paper_tools/merge_chain_corpora.py`. The resulting artefact `stage1_msc_train_s512.pt` contains 5928 chains (1899 PG-19 + 29 TV + 4000 MSC), $\sim$ 103k sessions, all eligible at `window_k`$\geq$3.